Before (W_O)

Suppose we have **4 heads**.

Sentence:

```
The boy kicked the ball because it was dirty.
```

Each head produces its **own output vector** for every token.

Example for the token "**ball**":

```
Head 1 → Grammar information

[0.2, 1.1]
```

```
Head 2 → Object relationship

[3.5, 0.7]
```

```
Head 3 → Color/description

[-0.8, 2.0]
```

```
Head 4 → Pronoun reference

[1.4, -0.3]
```

Step 1: Concatenate (Just Stick Together)

We simply join them.

```
Head1    Head2    Head3    Head4

[0.2,1.1 | 3.5,0.7 | -0.8,2.0 | 1.4,-0.3]
```

Nothing is merged yet.

It's literally:

```
Output =
Head1 output
+
Head2 output
+
Head3 output
+
Head4 output
```

Think of it like:

```
[Grammar | Objects | Description | Reference]
```

Each head still occupies its own "slot."

Problem

Suppose:

- Head 1 found useful grammar information.
- Head 4 found useful pronoun information.

What if the next layer needs **both together**?

Without processing, they're isolated:

```
[Grammar | Objects | Description | Reference]
```

No interaction.

Step 2: Apply (W_O)

Now we pass the concatenated vector through one learned matrix.

```
Concatenated Vector

↓

W_O
```

```
↓

Final Output Vector
```

Now information can mix.

Example (conceptually):

Instead of

```
Grammar = 5

Reference = 7
```

remaining separate,

(W_O) can learn something like

```
Final Feature =

0.6 × Grammar

+

0.4 × Reference
```

So the final vector contains **combinations of information from multiple heads**, not just the raw head outputs.

Analogy

Imagine four experts write reports.

```
Grammar Expert

↓

2 pages
```

```
Meaning Expert

↓

2 pages
```

```
2 pages
```

```
Reference Expert

↓

2 pages
```

```
Syntax Expert

↓

2 pages
```

Simply stapling them together isn't enough.

You hire an **editor**.

The editor:

- reads all reports,
- combines related information,
- removes redundancy,
- produces one coherent report.

That editor is **(W_O)**.

Entire Flow

```
Input

↓

Head 1
        \
Head 2 ----\
             > Concatenate
Head 3 ----/        │
        /           ▼
Head 4         W_O (mixes heads)
                    │
                    ▼
            Final Output
```

Short Answer (Interview Style)

> **How do we merge all heads?**

1. Each head computes its own attention output independently.
2. The outputs are **concatenated** into one larger vector (not summed or averaged).
3. A learned projection matrix **(W_O)** transforms this concatenated vector into a single output vector by **mixing information across heads**.
4. That final vector is passed to the next Transformer layer.

One-line memory trick:

- **Concatenation = stack the heads together.**
- **(W_O) = learn how to combine what each head discovered into one unified representation**